A per-run bound on the minimiser residual, tightened to the part of the model error the fit cannot absorb

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1 Notation

All quantities refer to one run's post-onset window unless stated. The table is complete; symbols not listed here do not appear.

τ	time since the identified onset, $\tau \in [0, \tau_{\text{end}}]$ [s]
τ_{end}	window length; $x = C_2\tau_{\text{end}}$ is its dimensionless form
$\tau_i,\ N$	the N sample instants $0 = \tau_1 < \cdots < \tau_N = \tau_{\text{end}}$ (logging period $T_s \approx 10$ ms)
y_i	measured roll/pitch rate at τ_i, from the flight IMU [rad/s]
$\omega_{\text{nom}}(\tau)$	nominal response $C_1(\cosh C_2\tau - 1) + C$ [rad/s]
$C_1,\ C_2,\ C$	amplitude $C_1 = K\dot{M}$, exponent $C_2 = \sqrt{W z_{CoM}/J_P}$, pre-onset baseline C
$K,\ \dot{M}$	calibrated gain $K = 1/(W z_{CoM})$ [1/(N m s)] and commanded ramp rate [N m/s]
$W,\ z_{CoM},\ J_P$	weight [N], CoM height [m], pivot-axis inertia [kg m^2]; $J_P = 1/(KC_2^2)$
$e_\omega(\tau)$	deviation of the true (noise-free) rate from ω_{nom} [rad/s]
n_i	in-window disturbance: everything in y that is neither ω_{nom} nor e_ω [rad/s]
u_i	regressor $u_i = \cosh C_2\tau_i - 1$; $\tilde{u}_i = u_i - \bar{u}$ its centred form
$V,\ A,\ P$	fitted family's span $V = \text{span}\{u, \mathbf{1}\}$; $A = [u\ \ \mathbf{1}]$; projector $P = A(A^\top A)^{-1}A^\top$, in closed form $P_{ij} = 1/N + \tilde{u}_i\tilde{u}_j/\sum_k \tilde{u}_k^2$
$\hat{r}$	minimiser residual $\hat{r} = (I - P)\,y$: the per-run fit with C_2 pinned and (C_1, C) free is exactly this projection
M	the fitted family $\{a\,u + c\,\mathbf{1} : a, c \in \mathbb{R}\}$, so that $\hat{r} = \min_{g\in M}$ in norm and M's span is V
$\hat{n}$	calibrated estimate of RMS(n), eq. (T.9); distinct from the envelope built on κ_b
$\rho(s)$	deviation forcing at time s: the gravity remainder $W z_{CoM}(\sin\varphi - \varphi)$ plus the ground-effect terms, evaluated on the true trajectory [N m]
$\bar{\rho}$	a priori pointwise bound $\|\rho(s)\| \le \bar{\rho}$ from the 10° box, as in (93) [N m]
k_j	kernel column: response of e_ω at the samples to a unit impulse of ρ at τ_j, i.e. $k_j(\tau_i) = \Delta s_j \cosh\big(C_2(\tau_i - \tau_j)\big)/J_P$ for $\tau_i \ge \tau_j$, else 0; Δs_j the trapezoidal quadrature weight [s]
Φ	a priori bound on RMS$\big((I-P)\,e_\omega\big)$, eq. (T.4) [rad/s, reported in °/s]
$E(\tau)$	the pointwise envelope of (93), $E = \bar{\rho}\sinh(C_2\tau)/(J_P C_2)$; note $E(\tau) = \bar{\rho}\sum_j \|k_j(\tau)\|$ — the same kernel without the projection
n_{hi}	the part of $\hat{r}$ above $f_c = 5$ Hz; attributable to disturbance because the family and e_ω are smooth on the window scale
κ	spectral shape ratio RMS($<f_c$)/RMS($>f_c$) of a disturbance record
κ_q	κ measured on the run's quiet stretch of equal length, ending at the excitation start
κ_{imp}	in-window implied ratio $\sqrt{\text{RMS}(\hat{r})^2 - \text{RMS}(n_{\text{hi}})^2}\,/\,\text{RMS}(n_{\text{hi}})$
$s_{\text{med}},\ \kappa_b$	median in-window enhancement $s_{\text{med}} = \text{med}(\kappa_{\text{imp}}/\kappa_q)$; campaign constant $\kappa_b = s_{\text{med}}\ \max_{\text{runs}} \kappa_q$
RMS(v)	$\sqrt{\frac{1}{N}\sum_i v_i^2}$ for sampled v (uniform grid; trapezoidal end weights where integrals are quoted)

2 The residual sees only what the family cannot absorb

The per-run fit holds C_2 at its physical value and minimises over (C_1, C), which is linear least squares on A: the fitted curve is Py and the residual is $\hat{r} = (I - P)y$. The record decomposes as $y = \omega_{\text{nom}} + e_\omega + n$, and $\omega_{\text{nom}} \in V$ exactly (it is $C_1 u + (C_1 \cdot 0 + C)\mathbf{1}$ evaluated at the same pinned C_2), so $(I - P)\,\omega_{\text{nom}} = 0$ and

$$\hat{r} \;=\; (I - P)\,(e_\omega + n). \tag{T.1}$$

This is Lemma 2 of the manuscript's Appendix C, eq. (C3), and its proof there ((C8)) is the annihilation just described; everything in this document builds on that identity, and the pointwise chain (C4)–(C6)

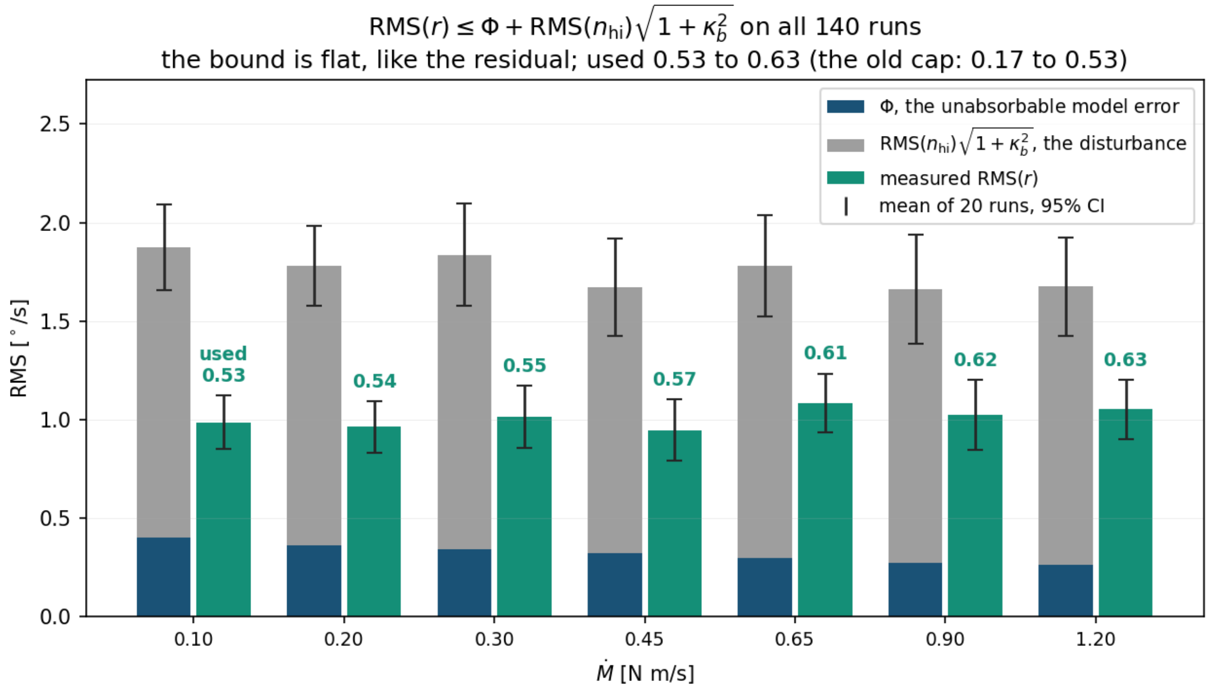

Figure 1: The bound (T.6) against the measured minimiser residual, means over the twenty runs at each rate with 95% t intervals. The left bar of each pair is the bound: the model part Φ of (T.4) at the bottom and the disturbance part $\mathrm{RMS}(n_{\mathrm{hi}})\sqrt{1+\kappa_b^2}$ — built from each run's own measured content above 5 Hz — stacked on top. The right bar is the measured residual, and the number above it is the fraction of the bound used. The bound holds on all 140 runs individually (per-run worst 0.82), and it is flat across the rates, 1.66 to 1.87°/s — like the residual, and unlike the old cap, which fell 5.77 to 1.98 with used fractions of 0.17 to 0.53.

with the constants λ_E and κ is retained unchanged. Where this document goes further is the *model* term: Appendix C's Lemma 3 charges the projection through the envelope, $D_i = \sum_j |P_{ij}|E_j$, while (T.4) below pushes the projection inside the Duhamel superposition, where the family annihilates the cosh part of every kernel column before the envelope is invoked at all. Hence

$$\mathrm{RMS}(\hat{r}) \;\le\; \mathrm{RMS}\big((I-P)e_\omega\big) + \mathrm{RMS}\big((I-P)n\big), \tag{T.2}$$

and this is where the old cap gave ground: it bounded the first term by $\mathrm{RMS}(e_\omega) \le \mathrm{RMS}(E)$, the size of the deviation itself. But the minimiser absorbs whatever part of e_ω lies in V, and most of it does. That is not an accident of the data — it is structural. e_ω is built from the kernel columns, and a shifted cosh splits as

$$\cosh\big(C_2(\tau-s)\big) = \cosh(C_2 s)\,\cosh(C_2\tau) - \sinh(C_2 s)\,\sinh(C_2\tau), \tag{T.3}$$

whose first part is $u + \mathbf{1} \in V$ *exactly*. What survives the projection is only the $\sinh(C_2\tau)$ part, weighted by $\sinh(C_2 s)$ — small for impulses early in the window — plus the truncation of the kernel before its impulse time. Norm-weighted over the columns, the family absorbs 68 to 91% of the kernel (79.7% in the median across runs) before any property of ρ is used.

Two objections deserve explicit answers, because both look fatal and neither is.

First: $P\,\omega_{\mathrm{nom}} = \omega_{\mathrm{nom}}$ *is an identity, not an approximation.* The nominal is $C_1 u + C\mathbf{1}$ built on the *same* pinned C_2 that builds V — both belong to the estimator — so the annihilation in (T.1) is exact by construction. The exponent that can genuinely differ is the *true* response's, and that discrepancy lives in e_ω by definition; it is priced in Sec. 3 below.

Second: the absorbed part has constant coefficients only per impulse, and that is all the argument needs. Aggregated, the Duhamel integral reads

$$e_\omega(\tau) \;=\; \cosh(C_2\tau)\,A(\tau) - \sinh(C_2\tau)\,B(\tau), \qquad A(\tau) = \tfrac{1}{J_P}\int_0^\tau \cosh(C_2 s)\rho(s)\,ds, \quad B(\tau) = \tfrac{1}{J_P}\int_0^\tau \sinh(C_2 s)\rho(s)\,ds,$$

and A, B vary with τ, so e_ω is *not* a fixed combination of cosh and sinh and is certainly not in V. But the projection is never applied to that aggregated form. It is applied per impulse, by linearity: in each column

k_j the shift s_j is fixed, the coefficients $\cosh(C_2 s_j)$ and $\sinh(C_2 s_j)$ in (T.3) are numbers, and the cosh part is annihilated column by column. The τ-dependence of A and B is precisely the running upper limit — the truncation indicator $\mathbf{1}_{\tau \geq s_j}$ of each column — and that truncation is carried, not discarded: it is the second surviving piece in $(I-P)k_j$, evaluated exactly in the numerical Φ. Measured on the exact nonlinear e_ω, the projection absorbs 61 to 67% of it in RMS across the rates, and the surviving 0.065–0.078°/s correlates −0.5 to −0.65 with $(I-P)\sinh(C_2\tau)$ — the sinh component and the truncation, exactly as the column split predicts.

3 The model term Φ

By Duhamel on the deviation dynamics $J_P \ddot{e}_\varphi = W z_{CoM}\, e_\varphi + \rho$ (with $e_\varphi(0) = \dot{e}_\varphi(0) = 0$; $e_\omega = \dot{e}_\varphi$), the sampled deviation is $e_\omega = \sum_j \rho(\tau_j)\, k_j$ to quadrature accuracy. Linearity of $I - P$, the triangle inequality applied pointwise, and $|\rho| \leq \bar{\rho}$ give

$$\big|\big[(I-P)\, e_\omega\big]_i\big| \;\leq\; \bar{\rho} \sum_j \big|\big[(I-P)\, k_j\big]_i\big| \qquad \Longrightarrow \qquad \mathrm{RMS}\big((I-P)e_\omega\big) \;\leq\; \Phi \;=\; \bar{\rho}\,\Big\| \textstyle\sum_j \big|(I-P)k_j\big| \Big\|_{\mathrm{RMS}}, \tag{T.4}$$

computable a priori from $(C_2, J_P, \bar{\rho})$ and the sample grid alone. Without the projection the same construction returns exactly the old envelope, $\bar{\rho}\sum_j |k_j(\tau)| = E(\tau)$, so the ratio $\mathrm{RMS}(E)/\Phi$ isolates what absorption is worth: a factor 12 at $\dot{M} = 0.10$ N m/s falling to 4 at 1.20, with $\Phi = 0.40$ down to 0.26°/s.

Both the representation and the bound were verified on the exact nonlinear tip-over: the kernel sum reconstructs e_ω to 3×10^{-4} °/s, and the realised $\mathrm{RMS}\big((I-P)e_\omega\big)$ is 0.065–0.078°/s across the rates — inside Φ with a factor 4 to 5 to spare, of which about 2 is $\bar{\rho}$ against the realised $|\rho|$ and about 2 is the sign coherence the triangle step discards.

If the pinned exponent mis-states the true one. Let the true response carry $C_2^* = (1+\varepsilon_2)C_2$. Whichever nominal one chooses to name, the correction is the same vector: with the calibrated nominal, the mismatch joins e_ω; with the true-exponent nominal, the annihilation fails by $(I-P)(\omega^* - \omega_{\mathrm{nom}})$ — and since $(I-P)\omega_{\mathrm{nom}} = 0$, both bookkeepings add identically

$$\Phi_{\varepsilon_2} \;=\; |\varepsilon_2|\, \big\|(I-P)\,\big[C_1 C_2 \tau \sinh(C_2\tau)\big]\big\|_{\mathrm{RMS}} + O(\varepsilon_2^2), \tag{T.4$'$}$$

the first-order response to a relative exponent error, projected. On the campaign grids this is 0.018–0.022°/s per 1% of ε_2 — 5–7% of Φ — flat across the rates: even $\varepsilon_2 = 5\%$ costs about 0.1°/s against a cap margin of 0.6–0.9. Since C_2 is itself fitted from the data by the stage-one PNLS with the exponent free, ε_2 at the few-percent level is what the calibration residuals allow, and the term is carried for completeness rather than out of need.

4 The noise term

Projections contract, so $\mathrm{RMS}\big((I-P)n\big) \leq \mathrm{RMS}(n)$, and the task is an estimate of $\mathrm{RMS}(n)$ that does not use the residual being bounded. Its high-frequency part needs no estimation: above $f_c = 5$ Hz the family and e_ω are smooth on the window scale and can contribute nothing — and by (T.4) their total contribution anywhere is at most Φ — so n_{hi} is measured directly from the record. The low-frequency part is extended from it by a spectral shape ratio: $\mathrm{RMS}(n) = \mathrm{RMS}(n_{\mathrm{hi}})\sqrt{1+\kappa^2}$ for the in-window κ.

The campaign fixes κ's scale but not its per-run value. On the 113 runs with a quiet stretch of equal window length, κ_q runs 0.15 (p10) to 1.31 (max), median 0.37, and, decisively, κ_q does not predict the in-window ratio run by run (Pearson $r = 0.04$, $p = 0.70$). A per-run noise model is therefore not available, and the bound uses the campaign envelope — measured entirely outside the windows it will be tested on:

$$\kappa_b \;=\; \max_{\mathrm{runs}} \kappa_q \;=\; 1.31, \qquad \sqrt{1+\kappa_b^2} \;=\; 1.65. \tag{T.5}$$

5 The bound, on the campaign

$$\boxed{\mathrm{RMS}(\hat{r}) \;\leq\; \Phi \;+\; \mathrm{RMS}(n_{\mathrm{hi}})\,\sqrt{1+\kappa_b^2}} \tag{T.6}$$

per run, with Φ from (T.4) and the run's own measured n_{hi}. On the 140 runs (means over the 20 at each rate):

$\dot{M}$ N m/s	Φ °/s	RMS(E) °/s	RMS(E)/Φ	cap (T.6) °/s	old cap °/s	RMS($\hat{r}$) °/s	used	old used
0.10	0.403	4.876	12	1.87	5.77	0.987	0.53	0.17
0.20	0.364	2.934	8	1.78	3.79	0.964	0.54	0.25
0.30	0.341	2.308	7	1.84	3.22	1.014	0.55	0.32
0.45	0.320	1.892	6	1.67	2.71	0.947	0.57	0.35
0.65	0.295	1.516	5	1.78	2.42	1.084	0.61	0.45
0.90	0.274	1.260	5	1.66	2.10	1.023	0.62	0.49
1.20	0.262	1.120	4	1.68	1.98	1.052	0.63	0.53

It holds on every run — per-run used fraction p10 0.49, median 0.56, p90 0.68, worst 0.82 — it is below the old cap at every rate, and it is *flat*: the cap runs 1.66 to 1.87°/s across a twelvefold change in ramp rate, matching the flat residual, where the old cap fell 5.77 to 1.98.

Little of the remaining width is removable within this form. The binding run fixes the smallest admissible cap factor at 1.28 (against the 1.65 in use), and even at that untenable limit — it is set by the very run it must cover — the mean used fraction only reaches 0.70. The distance from 0.58 to that ceiling is the price of covering the worst spectral shape in the campaign with one constant.

6 What the bound says, and what it cannot

Applying (T.4) to (T.1) two-sidedly gives

$$\left|\,\mathrm{RMS}(\hat{r}) - \mathrm{RMS}\big((I-P)n\big)\,\right| \;\le\; \Phi, \tag{T.7}$$

so the residual *is* the projected disturbance to within 0.26–0.40°/s. That is the flatness of Fig. 1 read as a theorem: the residual does not depend on $\dot{M}$ because the disturbance floor does not, and the model error, whose bound does fall with the rate, is capped at a third of the residual's size in the median — 11% of its power. The same statement fixes the limit of this check: a residual test can confirm ρ's contribution is below Φ, but it cannot resolve ρ inside the disturbance floor. The instrument that tests ρ is the forward check of (108), which propagates each perturbation along the measured trajectory and is not noise-limited.

7 The same bound by optimality alone, evaluated at the theory values

The two-line argument needs no projector at all:

$$\mathrm{RMS}(\hat{r}) \;=\; \min_{g\in M}\mathrm{RMS}(y-g) \;\le\; \mathrm{RMS}(e_\omega+n) \;\le\; \mathrm{RMS}(e_\omega)+\mathrm{RMS}(n), \tag{T.8}$$

valid because $\omega_{\text{nom}} \in M$ (same pinned C_2; freeing C_2 as well would only lower the left side). This is the old (VIII.3); what made its cap loose was never the argument but the numbers put into it. Putting in tight ones:

The model term at its theory value. Instead of the Chebyshev envelope RMS(E) = 4.88 down to 1.12°/s, take RMS(e_ω) itself from the forward-integrated exact tip-over — the same machinery the forward check of (108) rests on: 0.236 down to 0.165°/s across the rates.

The noise term at its calibrated estimate. Instead of the worst-case envelope of (T.5), take the middle of the distribution:

$$\hat{n} \;=\; \mathrm{RMS}(n_{\text{hi}})\sqrt{1+\big(s_{\text{med}}\,\kappa_q\big)^2}, \tag{T.9}$$

each run's own hi-band content, extended by its own quiet-window shape ratio times the campaign-median in-window enhancement (κ_q replaced by its campaign median 0.37 on the 27 runs without a quiet stretch).

The result (Fig. 2): the cap runs 1.20 to 1.30°/s against a measured 0.95 to 1.08 — used 0.76 to 0.85, covering the mean at every rate and 126 of 140 runs individually. The rest of the gap is not looseness to hunt down; it is the triangle step itself. Expanding the middle member of (T.8), $\mathrm{RMS}(e+n)^2 = \mathrm{RMS}(e)^2 + 2\langle e,n\rangle/N + \mathrm{RMS}(n)^2$, and the cross term averages to zero for a disturbance uncorrelated with the deviation,

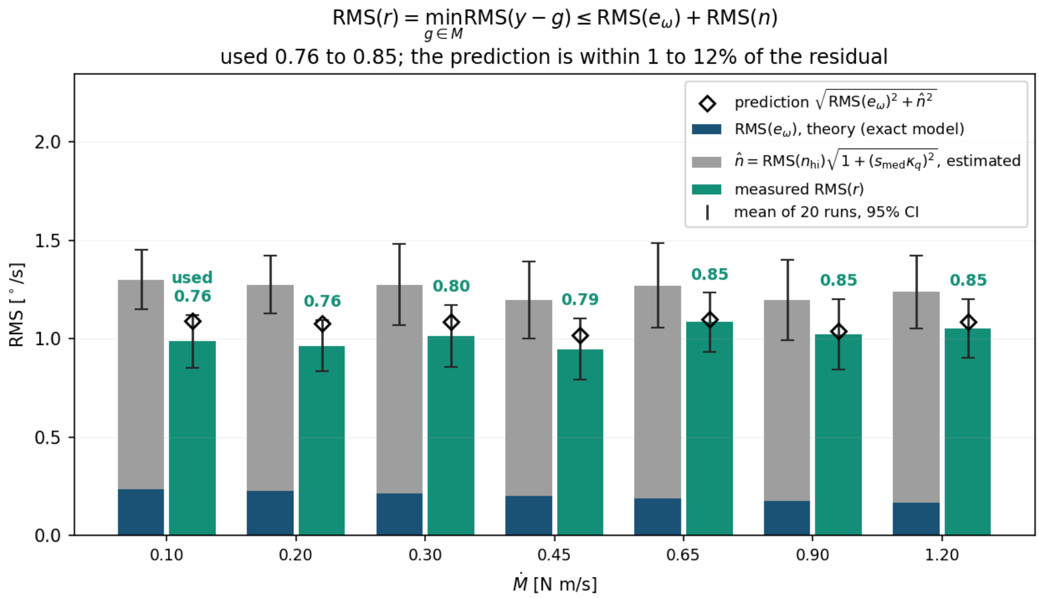

Figure 2: (T.8) with both terms at their theory values, means over the twenty runs per rate with 95% t intervals. The stacked bar is $\mathrm{RMS}(e_\omega)^{\text{theory}} + \hat{n}$; the green bar is the measured residual; the open diamond is the quadrature prediction $\sqrt{\mathrm{RMS}(e_\omega)^2 + \hat{n}^2}$, which lands within 1 to 12% of the residual at every rate. The used fraction is 0.76 to 0.85 and the mean cap covers the mean residual at all seven rates; per run it covers 126 of 140, the exceedances being the κ scatter that (T.9), an estimate, does not envelope.

so the *prediction* for the residual is the quadrature sum $\sqrt{\mathrm{RMS}(e_\omega)^2 + \hat{n}^2}$. Measured against the campaign it lands within 1 to 12% of the residual at every rate (the small excess being $\hat{n}$'s own margin and the two degrees of freedom the fit removes). The summed form exceeds it by exactly the triangle slack, at most $2\,\mathrm{RMS}(e)\,\mathrm{RMS}(n)$ inside the square, about 0.19°/s here.

The division of labour between the two routes is then clean. This section's evaluation is *nearly equal* to the data — used 0.76–0.85, prediction within 12% — because both terms sit at the middle of their distributions; the price is that it is an evaluation, and 14 runs exceed it. The κ_b/Φ version of (T.6) is the *guarantee* — it envelopes and holds on all 140 — at the price of the worst-case factor. One number cannot be both; the pair brackets the residual from the middle and from above.

8 Provenance of every constant

C_2, K, J_P are the per-configuration calibration constants, fixed before this analysis. $\bar{\rho}$ is a priori from the 10° box, as in (93). The kernel, the projector and hence Φ follow from those alone: the model part of (T.6) uses nothing measured in the window. The noise part uses one campaign scalar, $\max \kappa_q = 1.31$, from the quiet stretches only — the bound is built entirely from data outside the windows it is tested on.

One caveat is measured and stated rather than hidden. The in-window spectra are heavier at low frequency than the quiet ones — $s_{\text{med}} = \mathrm{med}(\kappa_{\text{imp}}/\kappa_q) = 1.60$, which is what pivoting on one gear edge adds — so on 5 of the 140 runs the residual exceeds the noise term of (T.6) alone, and it is Φ's margin that covers them. A reader who wants the noise term safe by construction can take $\kappa_b = s_{\text{med}} \times 1.31 = 2.09$: that variant also holds on all 140 runs, at a mean used fraction of 0.44, and prices the in-window enhancement at about 0.6°/s of cap. The tight version is reported as the result and the conservative one as the price of removing its caveat.